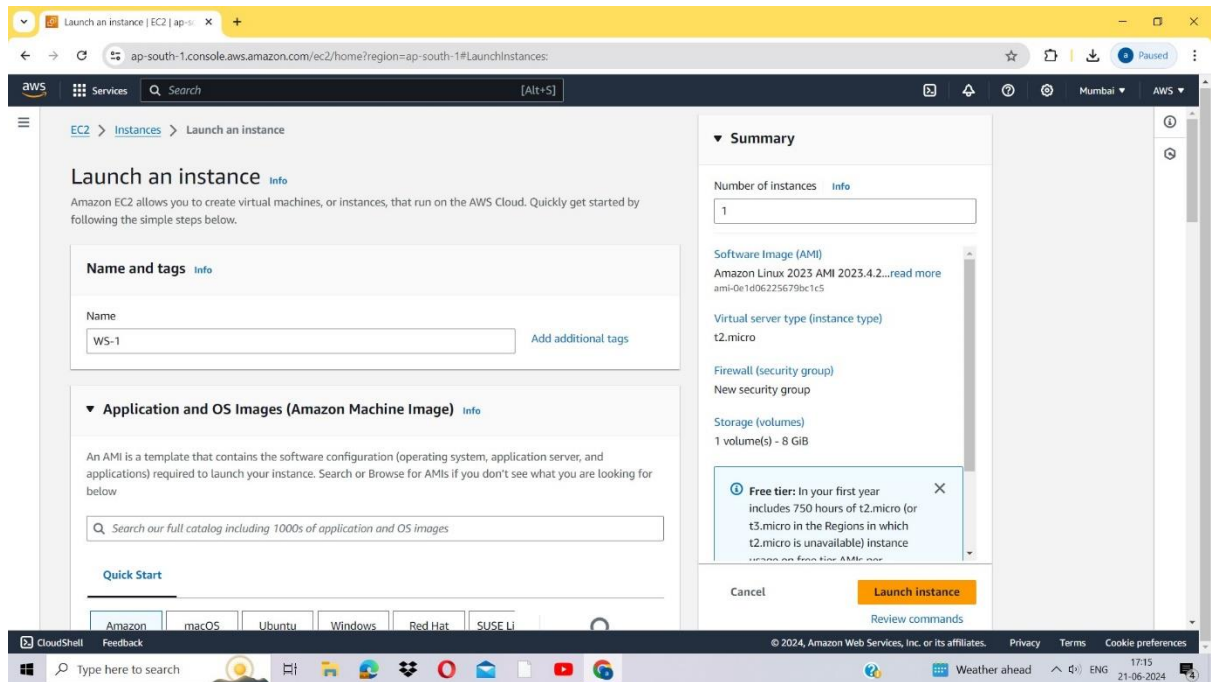


Elastic Load Balancer

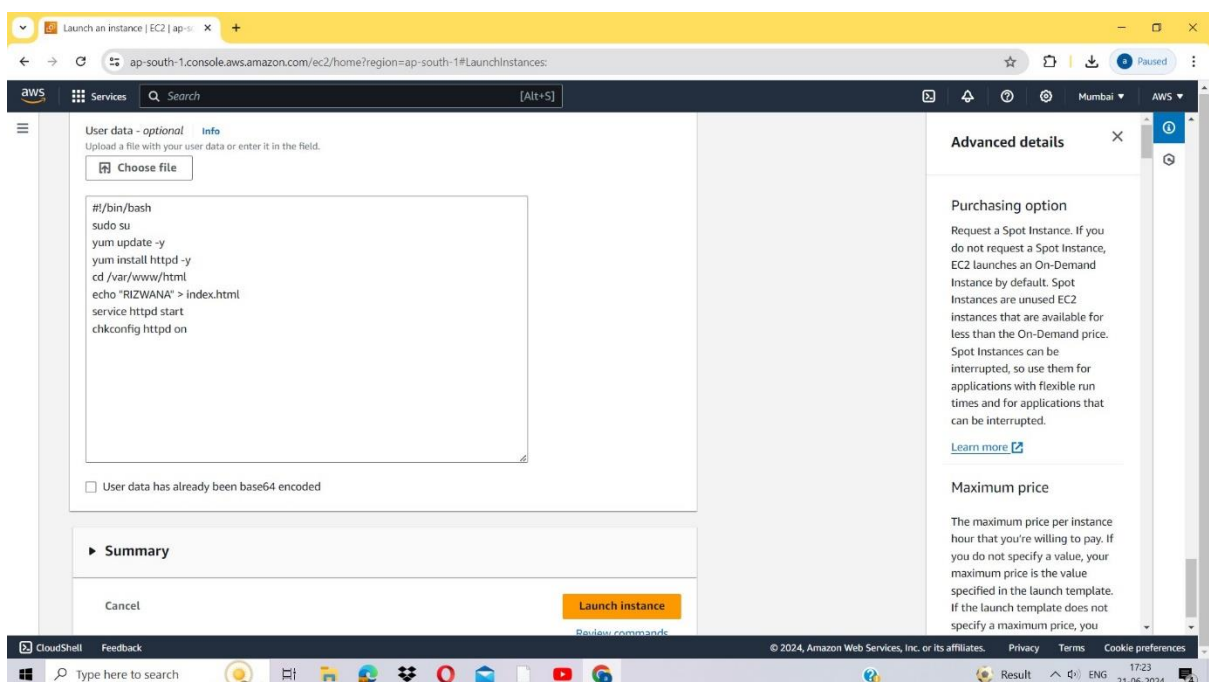
Name:Vempalli Rizwana Rollno:21HM1A05G0 Email id:rrizzu139@gmail.com

Steps to configure an Application load balancer in AWS

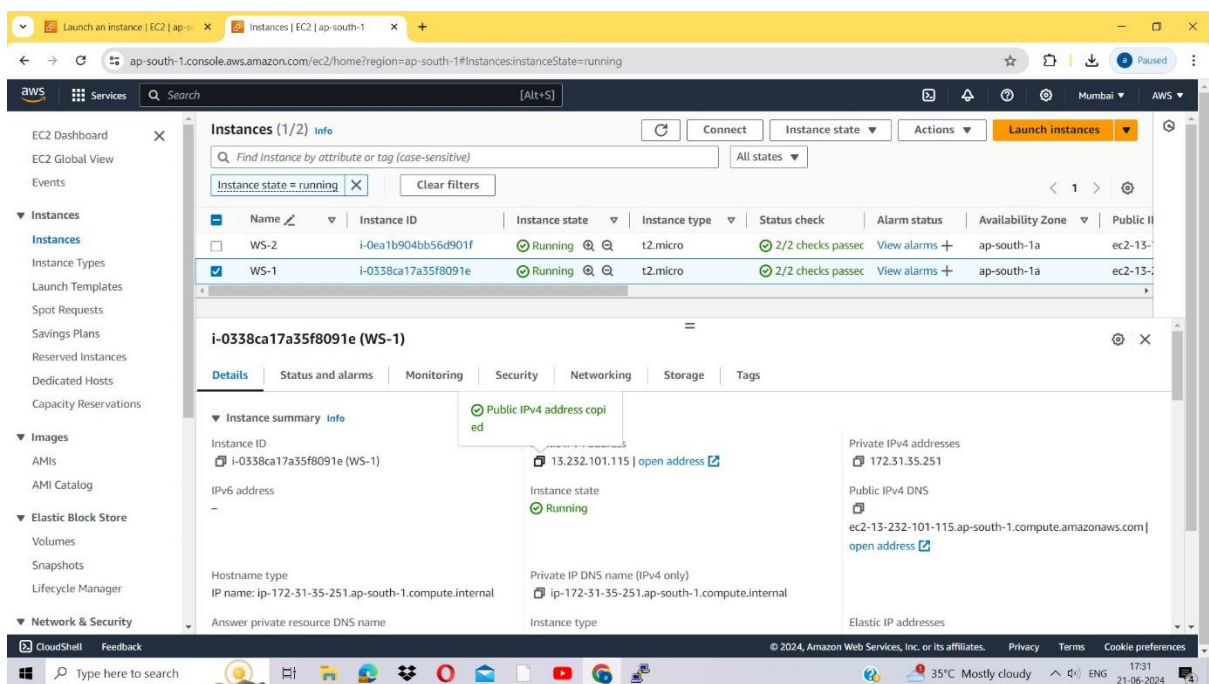
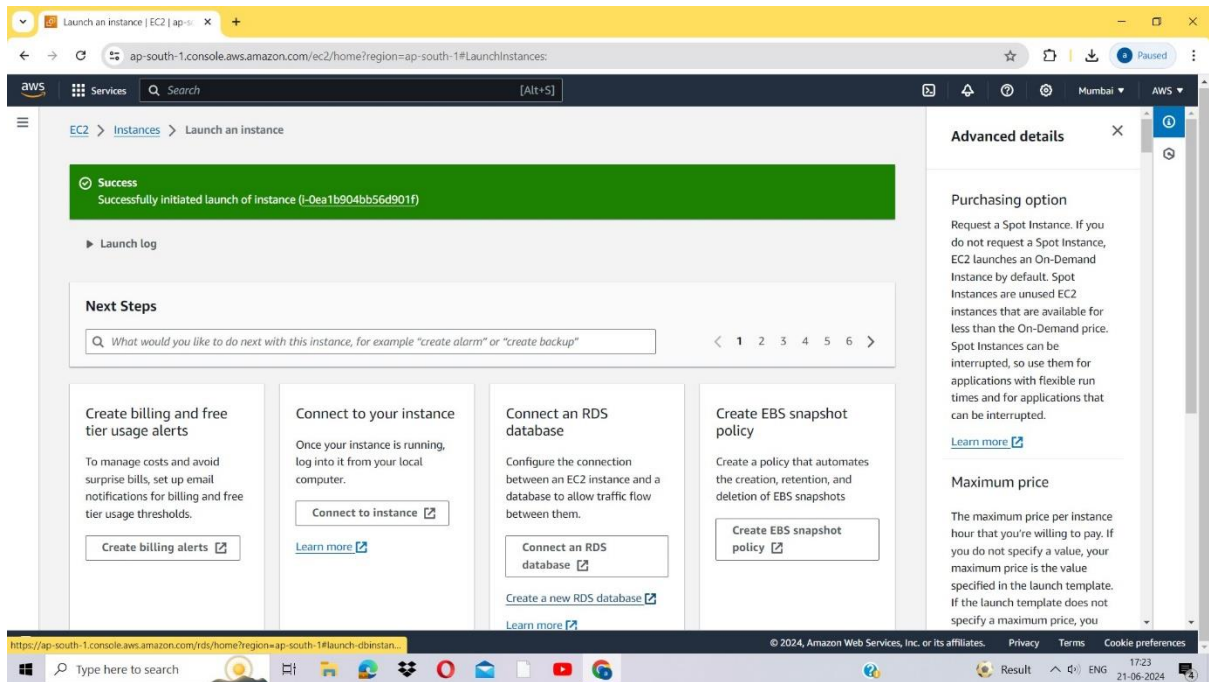
- Launch the two instances on the AWS management console named ws-1 and ws-2. Go to services and select the load balancer.



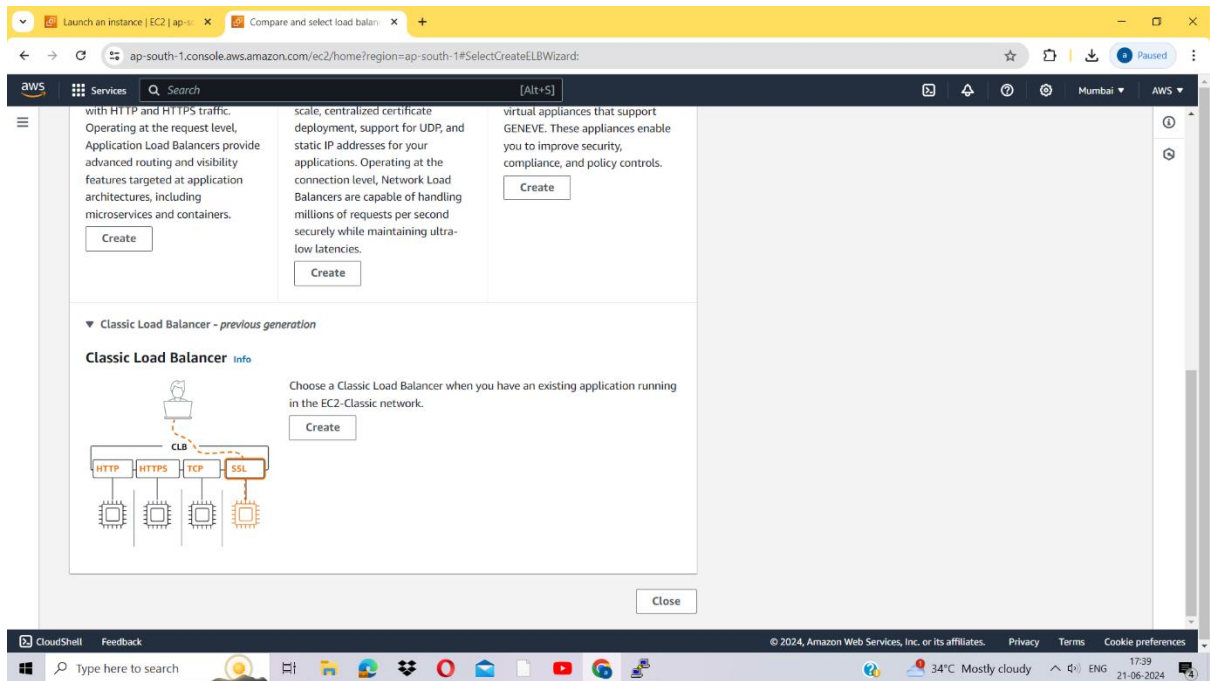
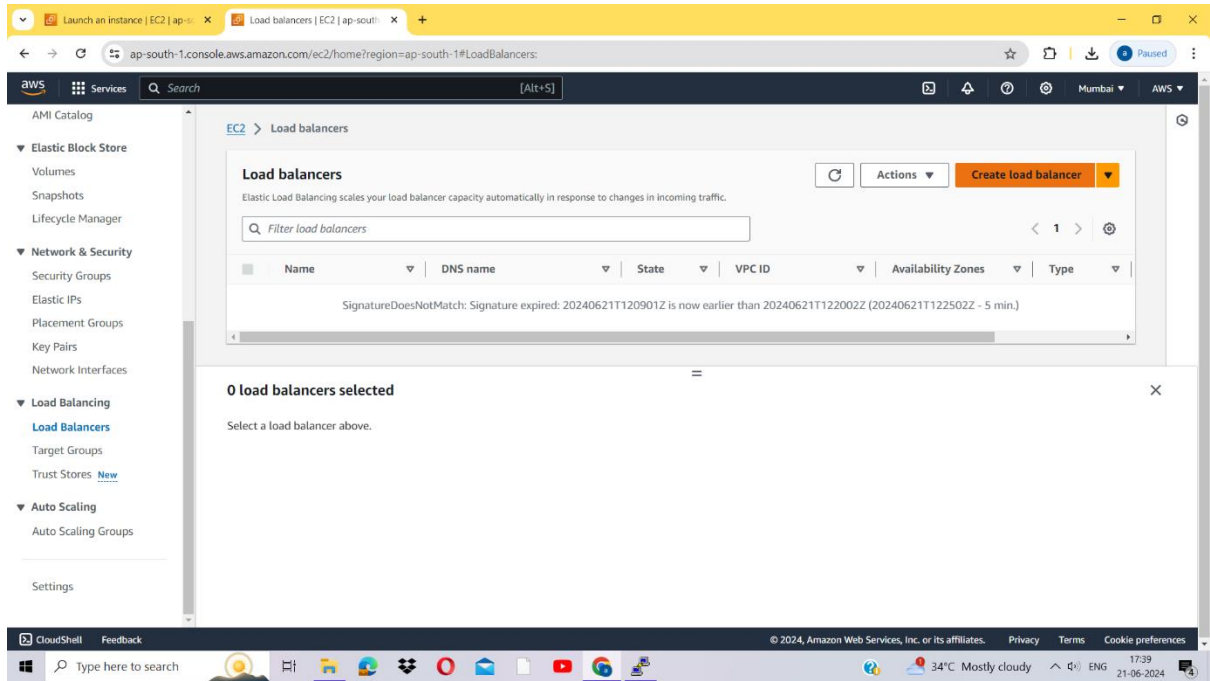
- Then click Launch instance.

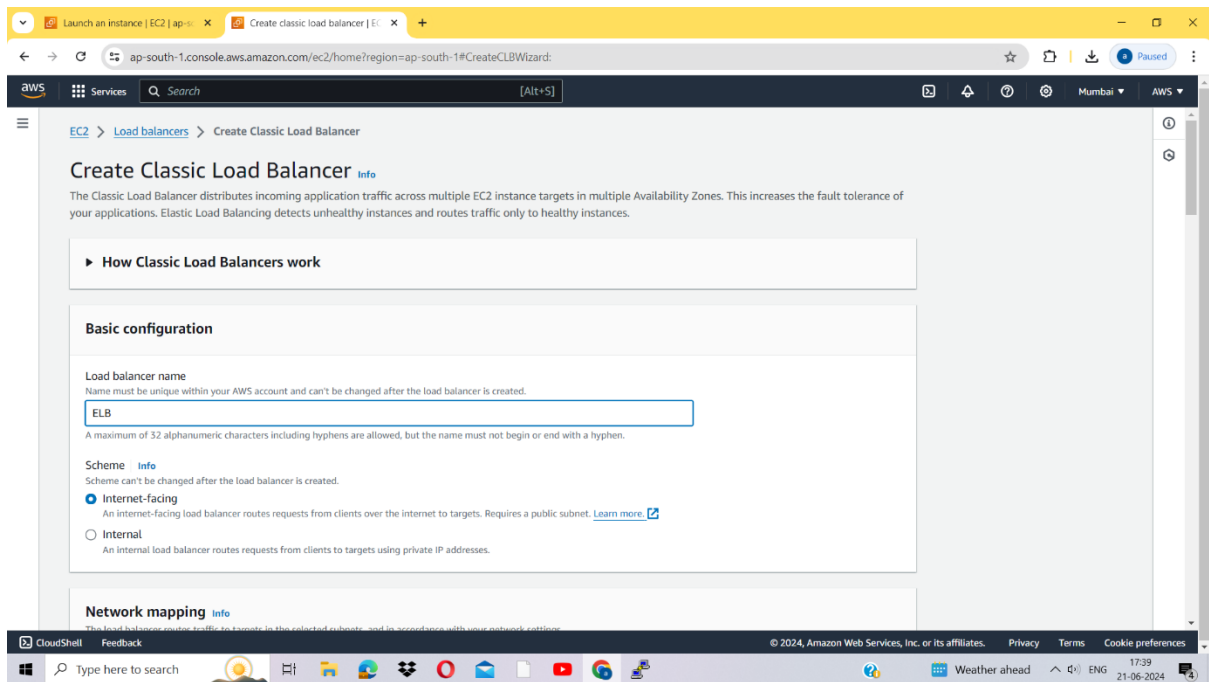


- Congratulations on successfully launching an instance

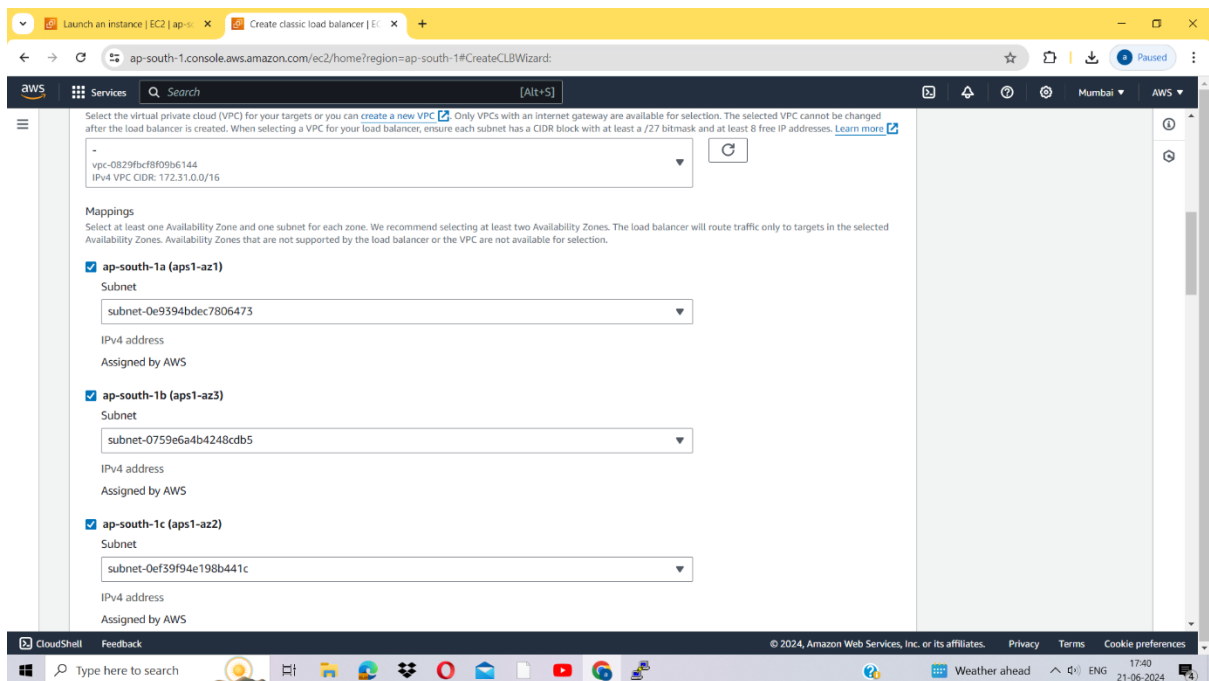


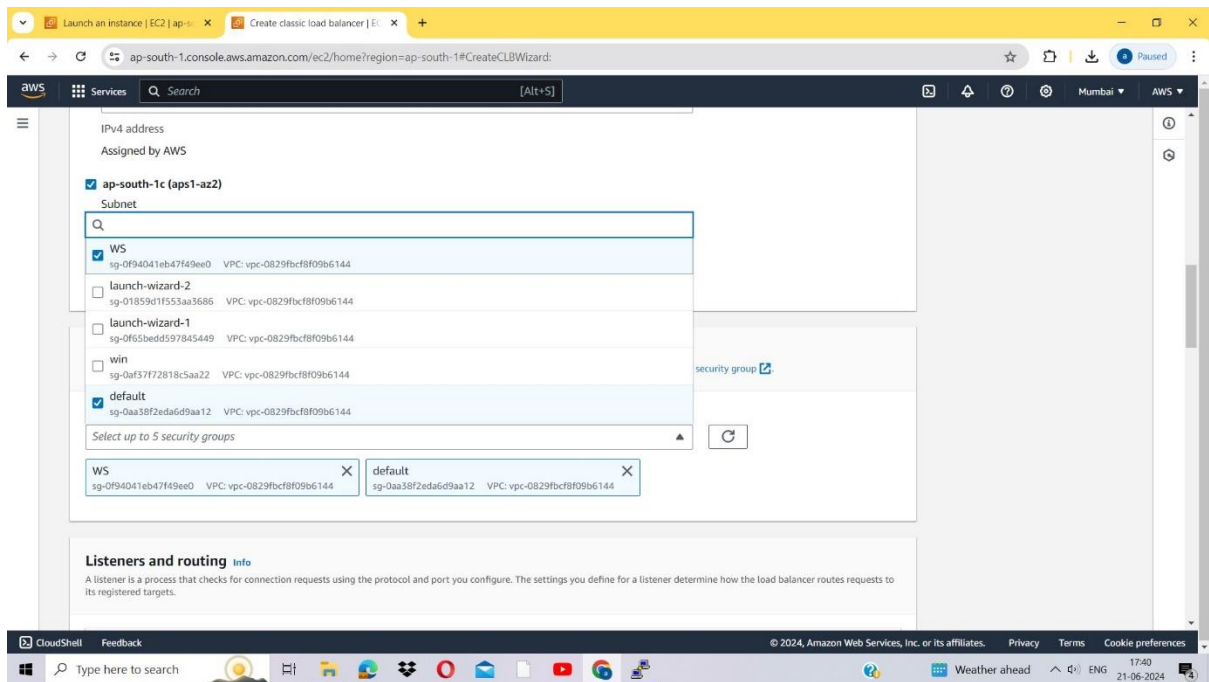
Step 2: Click on Create the load balancer.



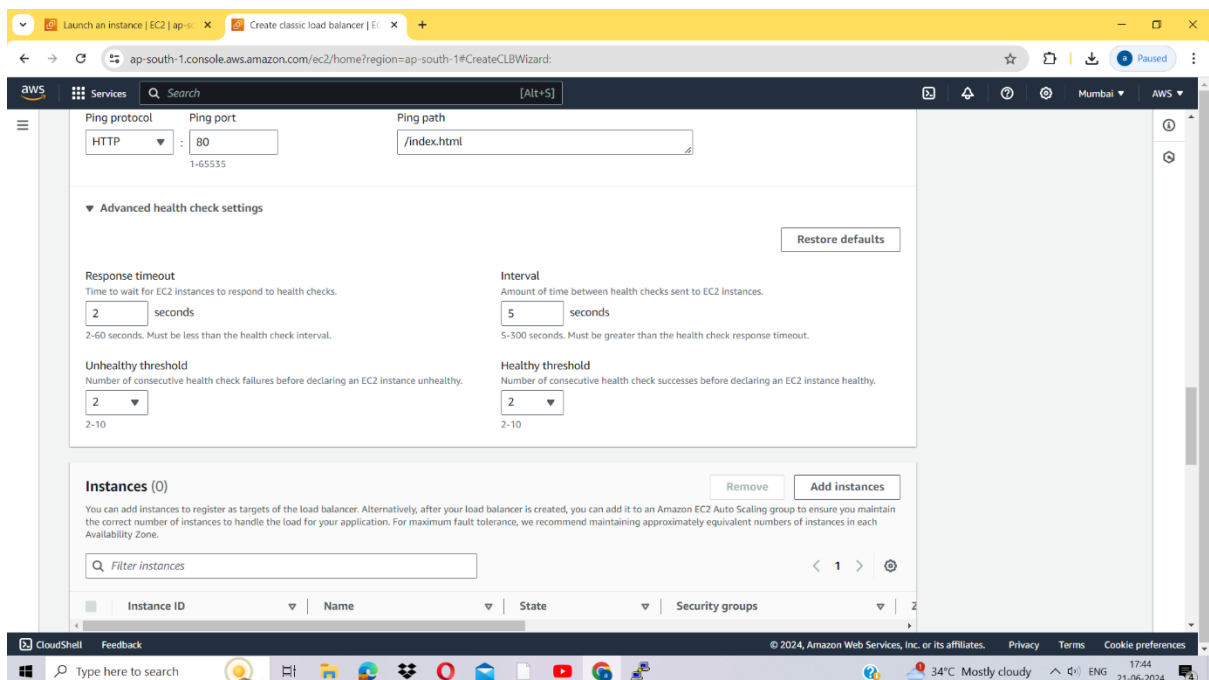


- Add at least 2 availability zones. Select us-east-1a and us-east-1b

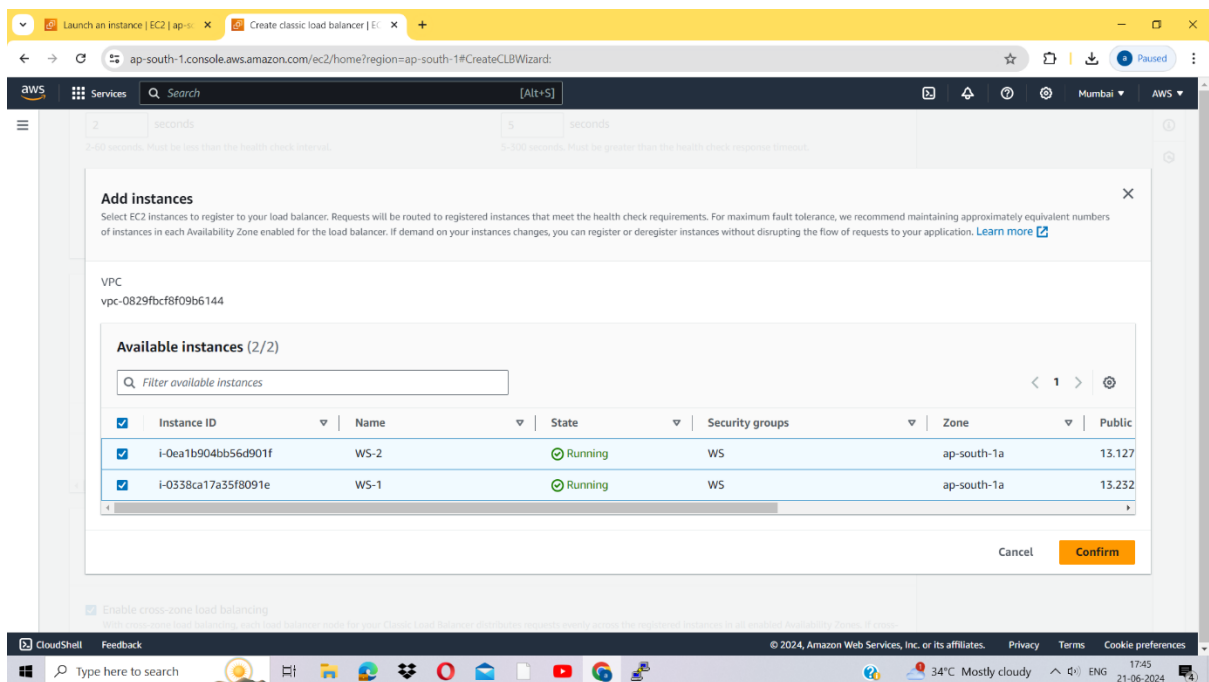




- Monitor the health of registered targets using the configured health check settings.

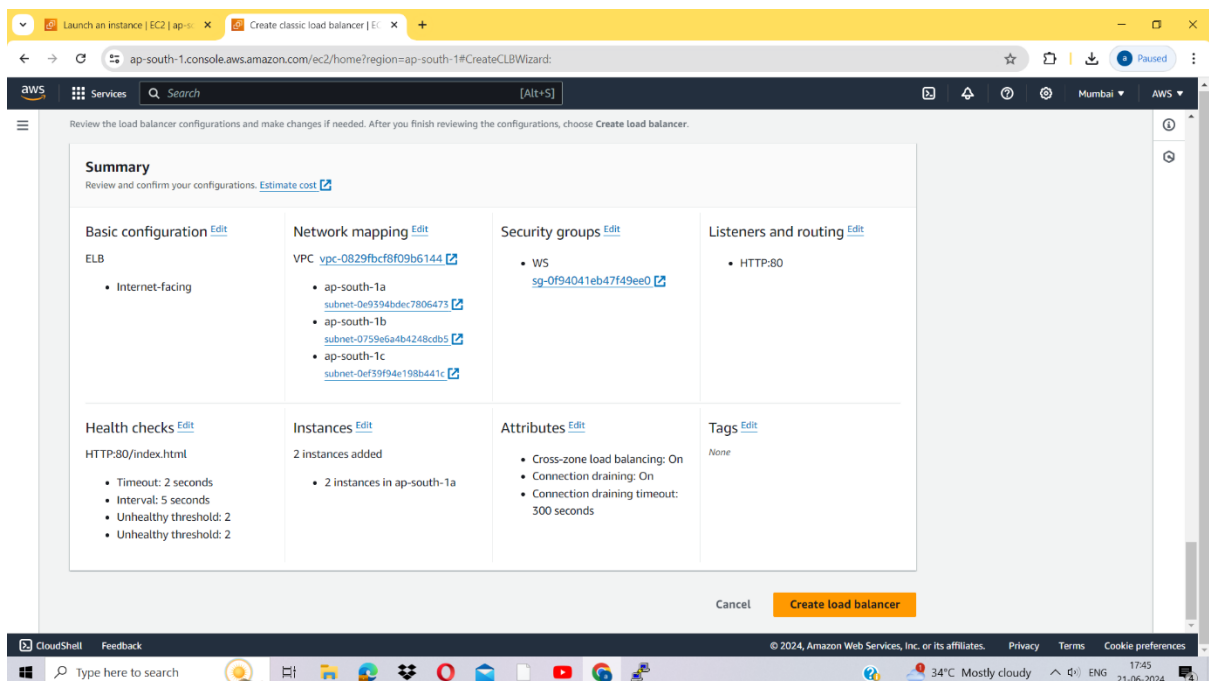


Choose WS-1 and WS-2 and click on Add to register.
Click on Next: Review.



Click on confirm

And then create load balancer.



Congratulations!! You have successfully created a load balancer.

Click on close

Load balancers | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancers:loadBalancerArn=my-elb;search=my-elb

EC2 Dashboard

EC2 Global View

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33°C Mostly cloudy 18:13 21-06-2024

Successfully created load balancer: my-elb

It might take a few minutes for your load balancer to be fully set up and ready to route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

EC2 > Load balancers > my-elb

Load balancers (1)

Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers

1 match

my-elb

Clear filters

0 load balancers selected

Select a load balancer above.

ModifyInboundSecurityGroupRules

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#ModifyInboundSecurityGroupRules:securityGroupId=sg-0f94041eb47f49ee0

EC2 > Security Groups > sg-0f94041eb47f49ee0 - WS > Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-017735092af52bc84	SSH	TCP	22	Custom	
-	HTTP	TCP	80	Anywh...	

Add rule

Rules with source of 0.0.0.0/0 or :::0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Cancel Preview changes Save rules

Load balancers | EC2 | ap-south-1

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#LoadBalancers:search=my-elbv=3.\$case=tags:false%5C,client:false,\$regex=tags:false%5C,client:false

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CloudShell Feedback

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Type here to search

Load balancers (1/1)
Elastic Load Balancing scales your load balancer capacity automatically in response to changes in incoming traffic.

Filter load balancers 1 match

my-elb X Clear filters

☒	Name	DNS name	State	VPC ID	Availability Zones	Type
☒	my-elb	my-elb-135010438.ap-sou...	-	vpc-0829fbcf8f09b6144	3 Availability Zones	classic

Load balancer: my-elb

south-1c (aps1-az2)

DNS name copied

my-elb-135010438.ap-south-1.elb.amazonaws.com (A Record)

This Classic Load Balancer can be migrated to a next generation load balancer. Migration wizard uses your load balancer's current configurations to create a new load balancer. [Learn more.](#)

Launch migration wizard X

► Distribution of targets by Availability Zone (AZ)
For each enabled Availability Zone, you can view the number of registered instances and their current health states. Selecting any values here will apply the corresponding filter to the Target instances table.

Load balancers | EC2 | ap-south-1

New Tab

my-elb-135010438.ap-south-1.elb.amazonaws.com

my-elb-135010438.ap-south-1.elb.amazonaws.com

my-elb-135010438.ap-south-1.elb.amazonaws.com - Google Search

Images

Google

Search Google or type a URL

Add shortcut

Customize Chrome

Type here to search

Hot weather

18:20
21-06-2024

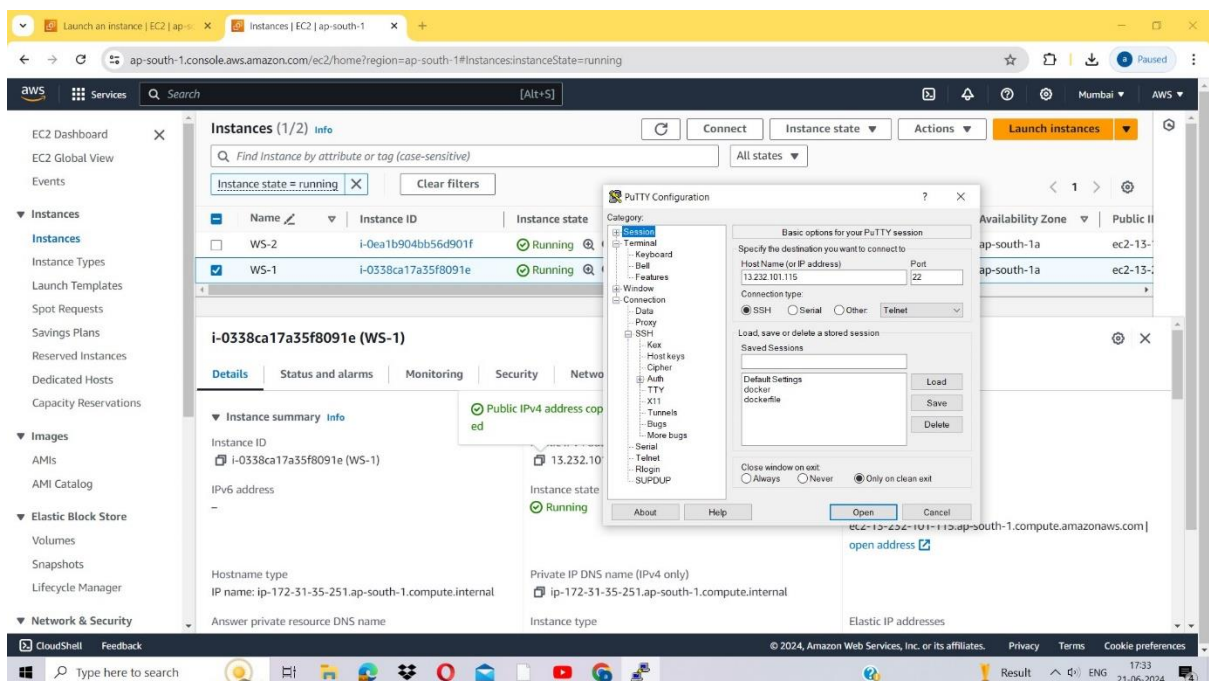


RIZWANA



Open up PuTTY and in the 'Host Name (or IP address)' input box, paste the IP.

Select the connection type as SSH and port as 22.



- A login page will open. Enter ec2-user (it's the default user created by EC2 while launching)
- If everything goes well you should see a Linux prompt where you can run server commands.

```

root@ip-172-31-35-251/home/ec2-user
login as: ec2-user
Authenticating with public key "imported-openssh-key"

      _ _ _ _ _
     _ _ _ _ _
    _ _ _ _ _
   _ _ _ _ _
  _ _ _ _ _
 _ _ _ _ _
_ _ _ _ _

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

Last login: Fri Jun 21 12:20:17 2024 from 223.165.51.23
[ec2-user@ip-172-31-35-251 ~]$ sudo su
[root@ip-172-31-35-251 ec2-user]# yum update -y
Last metadata expiration check: 0:56:36 ago on Fri Jun 21 12:03:23 2024.
Dependencies resolved.
Nothing to do.
Complete!
[root@ip-172-31-35-251 ec2-user]# yum install httpd -y
Last metadata expiration check: 0:56:51 ago on Fri Jun 21 12:03:23 2024.
Dependencies resolved.
=====
Package           Arch      Version      Repository    Size
-----
Installing:
httpd             x86_64    2.4.59-2.amzn2023      amazonlinux   47 k
Installing dependencies:
apr               x86_64    1.7.2-2.amzn2023.0.2    amazonlinux   129 k
apr-util          x86_64    1.6.3-1.amzn2023.0.1    amazonlinux   98 k
generic-logos-httpd  noarch    18.0.0-12.amzn2023.0.3  amazonlinux   19 k
httpd-core        x86_64    2.4.59-2.amzn2023      amazonlinux   1.4 M
httpd-filesystem  noarch    2.4.59-2.amzn2023      amazonlinux   14 k
httpd-tools       x86_64    2.4.59-2.amzn2023      amazonlinux   81 k
libbrotli         x86_64    1.0.9-4.amzn2023.0.2    amazonlinux   315 k
mailcap           noarch    2.1.49-3.amzn2023.0.3   amazonlinux   33 k
Installing weak dependencies:
apr-util-openssl  x86_64    1.6.3-1.amzn2023.0.1    amazonlinux   17 k
mod_http2         x86_64    2.0.27-1.amzn2023.0.2    amazonlinux   166 k
mod_lua           x86_64    2.4.59-2.amzn2023      amazonlinux   61 k
Transaction Summary
-----
Install 12 Packages

Total download size: 2.3 M
Installed size: 6.9 M
Downloading Packages:
(1/12): apr-1.7.2-2.amzn2023.0.2.x86_64.rpm    2.0 MB/s | 129 kB    00:00
(2/12): apr-util-openssl-1.6.3-1.amzn2023.0.1.x 251 kB/s | 17 kB    00:00

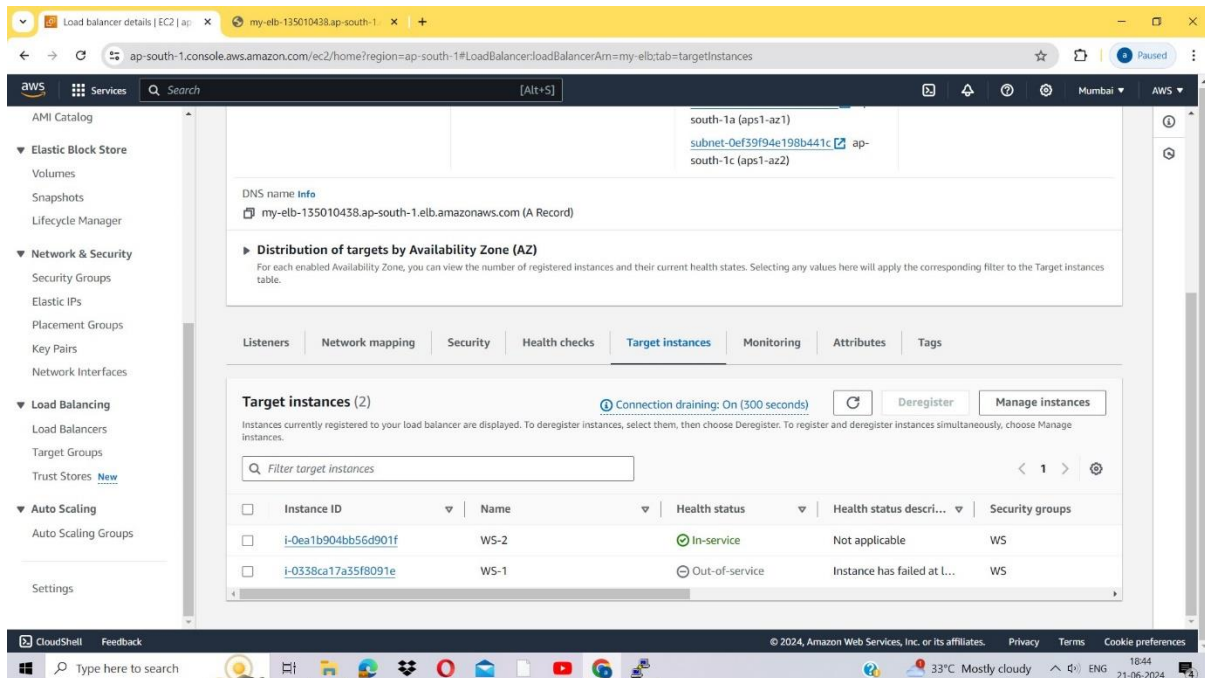
```

```
root@ip-172-31-35-251:/home/ec2-user
Installing      : httpd-core-2.4.59-2.amzn2023.x86_64      6/12
Installing      : mod_httpd-2.0.27-1.amzn2023.0.2.x86_64  9/12
Installing      : mod_lua-2.4.59-2.amzn2023.x86_64        10/12
Installing      : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 11/12
Installing      : httpd-2.4.59-2.amzn2023.x86_64          12/12
Running scriptlet: httpd-2.4.59-2.amzn2023.x86_64          12/12
Verifying       : apr-1.7.2-2.amzn2023.0.2.x86_64         1/12
Verifying       : apr-util-1.6.3-1.amzn2023.0.1.x86_64     2/12
Verifying       : apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64 3/12
Verifying       : generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch 4/12
Verifying       : httpd-2.4.59-2.amzn2023.x86_64           5/12
Verifying       : httpd-core-2.4.59-2.amzn2023.x86_64      6/12
Verifying       : httpd-filesystem-2.4.59-2.amzn2023.noarch 7/12
Verifying       : httpd-tools-2.4.59-2.amzn2023.x86_64     8/12
Verifying       : libbrotli-1.0.9-4.amzn2023.0.2.x86_64    9/12
Verifying       : mailcap-2.1.49-3.amzn2023.0.3.noarch     10/12
Verifying       : mod_httpd-2.0.27-1.amzn2023.0.2.x86_64   11/12
Verifying       : mod_lua-2.4.59-2.amzn2023.x86_64         12/12

Installed:
apr-1.7.2-2.amzn2023.0.2.x86_64
apr-util-1.6.3-1.amzn2023.0.1.x86_64
apr-util-openssl-1.6.3-1.amzn2023.0.1.x86_64
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch
httpd-2.4.59-2.amzn2023.x86_64
httpd-core-2.4.59-2.amzn2023.x86_64
httpd-filesystem-2.4.59-2.amzn2023.noarch
httpd-tools-2.4.59-2.amzn2023.x86_64
libbrotli-1.0.9-4.amzn2023.0.2.x86_64
mailcap-2.1.49-3.amzn2023.0.3.noarch
mod_httpd-2.0.27-1.amzn2023.0.2.x86_64
mod_lua-2.4.59-2.amzn2023.x86_64

Complete!
[root@ip-172-31-35-251 ec2-user]# echo "ELB" > index.html
[root@ip-172-31-35-251 ec2-user]# ls
index.html
[root@ip-172-31-35-251 ec2-user]# pwd
/home/ec2-user
[root@ip-172-31-35-251 ec2-user]# service httpd start
Redirecting to /bin/systemctl start httpd.service
[root@ip-172-31-35-251 ec2-user]# chkconfig httpd on
Note: Forwarding request to 'systemctl enable httpd.service'.
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service - /usr/lib/systemd/system/httpd.service.
[root@ip-172-31-35-251 ec2-user]# cat index.html
ELB
[root@ip-172-31-35-251 ec2-user]# rm index.html
rm: remove regular file 'index.html'? y
[root@ip-172-31-35-251 ec2-user]# ls
```

- In Linux, the rm command is used to remove files and directories



Conclusion

Successfully created an Elastic Load Balancer and prove its high availability, automatic scaling, and robust security features.

